

Results (Supplementary Figure S4)

Acute inhibition is not mitigated by divalent cation supplementation in rodents

Previous reports have found divalent cation supplementation in ASO or siRNA dosing solutions can mitigate acute responses, but it is unclear whether activation or inhibition phenotypes are affected [17, 18, 21]. Here, we investigated how acute inhibition is affected by divalent cations (Ca^{2+} or Mg^{2+}) in the dosing solution of ASOs delivered via ICV injection in mice. Animals were treated with aCSF containing 1-100 mM Ca^{2+} or Mg^{2+} without or with an ASO known to cause an acute inhibition response (ASO6). Animals were observed from 0-3 hours post-dose, and acute inhibition was scored at 3 hours post-dosing. No acute inhibition response was observed with the vehicle alone and the acute inhibition response was not altered with either Ca^{2+} or Mg^{2+} supplementation (Supplementary Figure S4A-B), in mice. Of note, all animals treated with 100 mM Ca^{2+} died immediately at dosing and were not included in the 3 hour post-dose evaluation (Supplementary Figure S4A-B). Of note, the solution was not controlled for tonicity and which may contribute to the effect of high IT Ca^{2+} alone. These results indicate that the acute inhibition response in rodents described here is not mitigated by divalent cation supplementation. However, extremely high levels of Ca^{2+} can be lethal when delivered to the CSF and may confound interpretation of acute tolerability studies with ASOs or siRNAs.

Supplementary Materials and Methods

Grip strength and open field assessments of motor function in mice

Mice were acclimated to the testing room for a minimum of 60 minutes prior to each round of behavioral testing. Grip strength: The Chatillon-Ametek Digital Force Gauge, DFE II (Columbus Instruments, Columbus, OH) was used to determine the strength exerted by the hindlimbs of each animal in response to a constant downward force. Data presented as average of 4 trials as grip force (Gf) normalized to body weight (Gf/g). Open field: Kinder Scientific Open Field Arenas with 16 x 16 beams located 1 inch apart were used for this test. Mice were placed individually into the center of the arena and total activity was measured as the number of beam breaks over 300 seconds.

Supplementary Figure 1

A Acute Inhibition scoring following IT administration in rodents

3h Acute Inh. score	Rodent IT injection Acute Inhibition scale
0	Bright, alert, responsive
1	No tone/movement in tail
2	Weak posterior posture
3	Hind limbs don't support weight, but can still move
4	Hind paws don't move – full hindlimb paralysis
5	Weak anterior posture
6	Fore paws don't move, but animal is still breathing
7	Death

Note: Rats are observed 3 hours after dosing and a score is assigned.

B Acute Inhibition scoring following ICV administration in rodents

3h Acute Inh. score	Rodent ICV injection Acute Inhibition scale
0	Bright, alert, responsive
1	Gait slightly impaired
2	Hunched and/or impaired gait
3	Forward movement only after lift
4	Any movement after lift
5	Lateral recumbency with response to tail pinch
6	No response to tail pinch but still breathing
7	Death

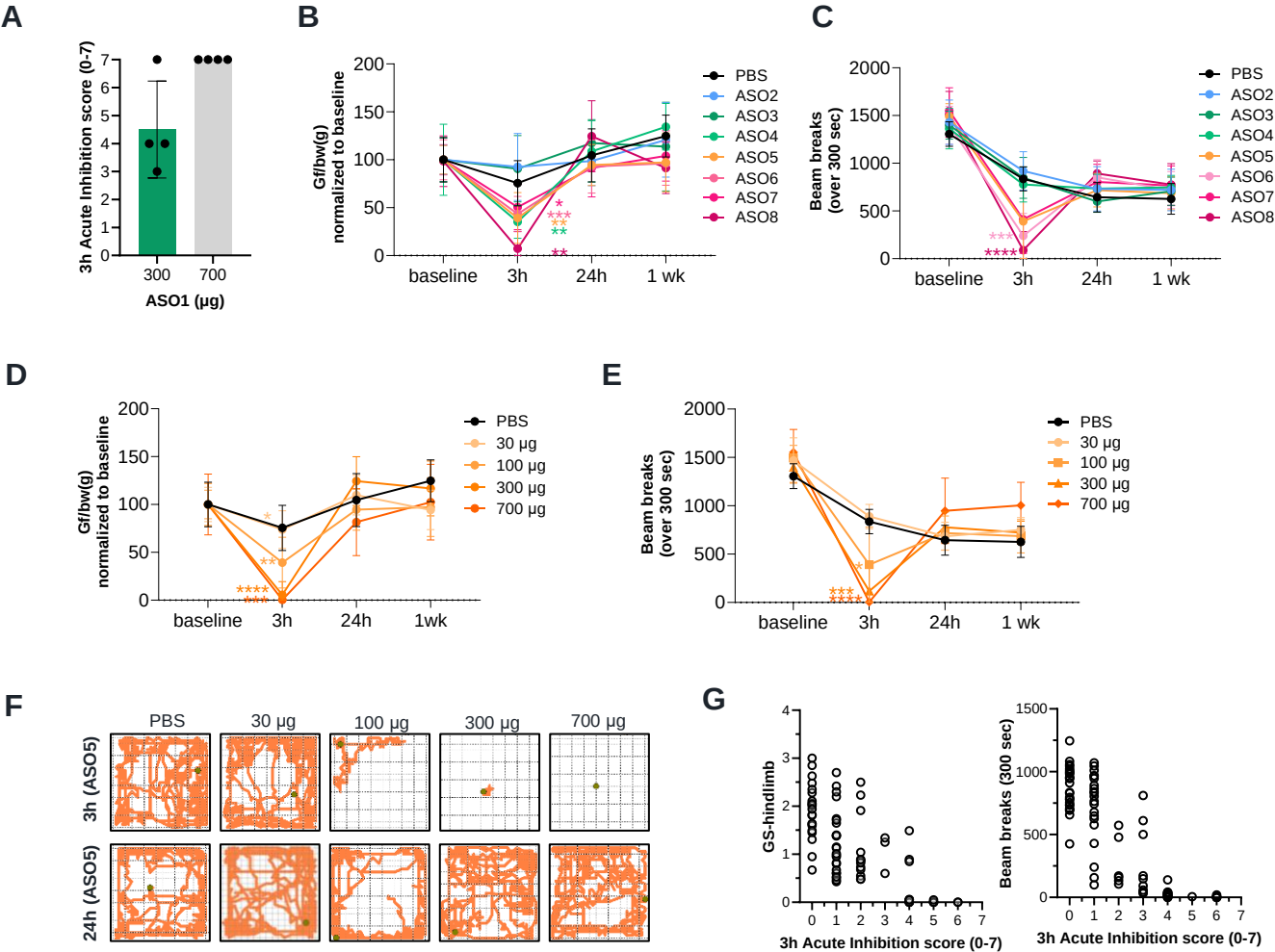
Note: Mice or rats are observed 3 hours after dosing and a score is assigned.

C Acute Inhibition scoring following IT administration in NHP

NHP Acute Inh. score	Exam	Observation
2	Knee Jerk Reflex	The quick extension of a hind leg is noted following the tapping of the patellar ligament with a hard object: absence of response = + 1 pt per side
2	Cutaneous Reflex	The reaction to a series of needle pricks along each side of the abdomen is recorded: absence of response = + 1 pt per side
2	Proprioceptive Reflex	Each foot is inverted and the ability to right the foot is noted: absence of response = + 1 pt per side
2	Sensory Foot Reflex	The foot is stimulated with a sharp object and an avoidance response is observed: absence of response = + 1 pt per side
1	Tail Reflex	When the tail is touched with a sharp object, the tail movement to avoid the stimulus is noted: absence of response = +1 pt
9	Total score (cumulative)	

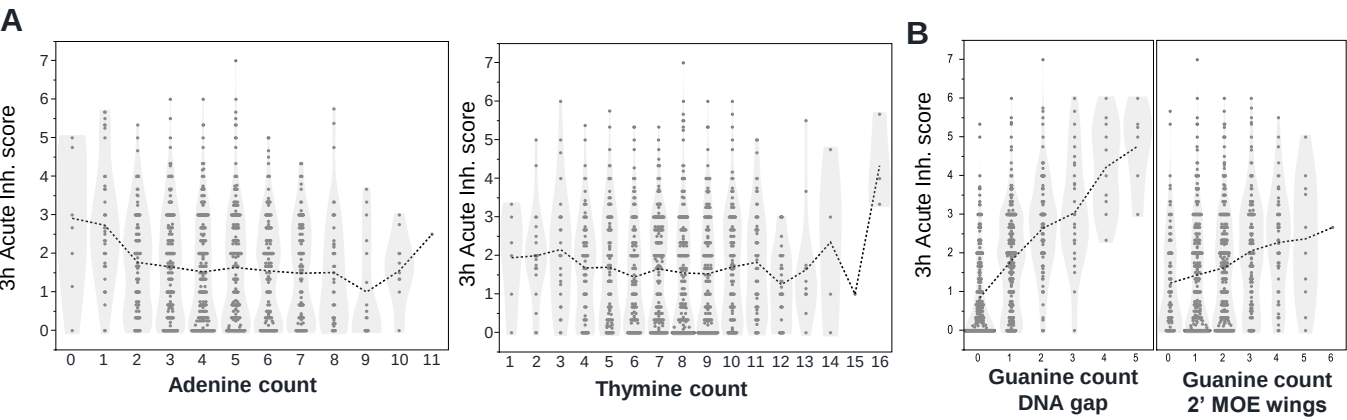
Supplementary Figure 1. Acute Inhibition scales for IT or ICV administration in rodents, or IT administration in NHP. Acute Inhibition scales for IT (A) or ICV (B) administration in rodents and IT administration in NHP (C).

Supplementary Figure 2



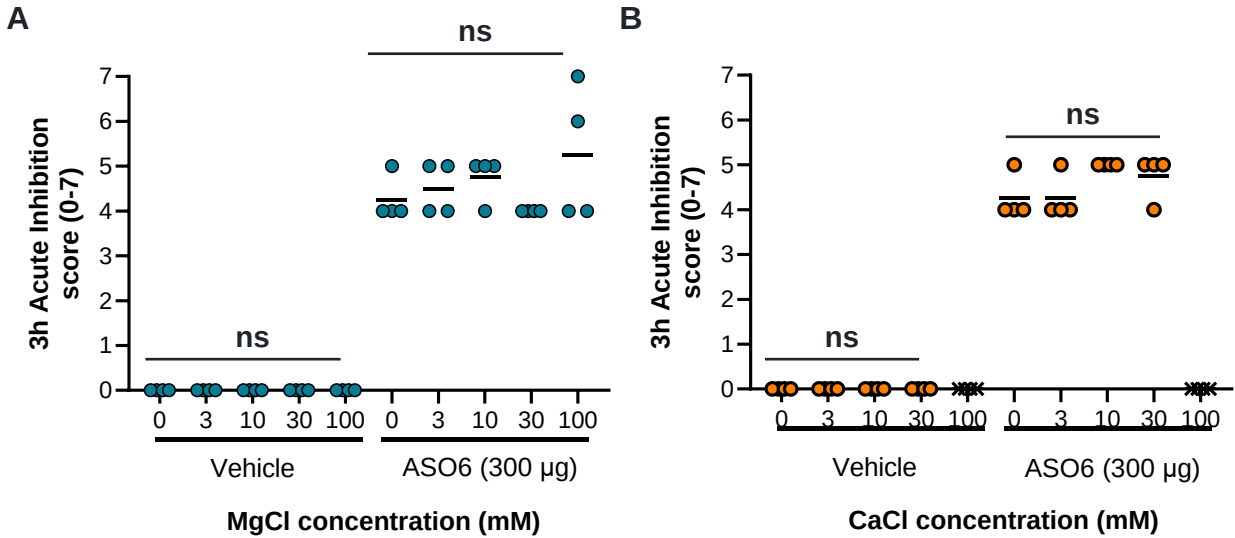
Supplementary Figure 2. Acute inhibition is sequence dependent and translates to mouse, where measures of grip strength and open field activity correlate with 3h Acute Inhibition scores. **(A)** Acute Inhibition scores 3 hours post-ICV injection of 300 or 700 µg ASO1 in mice as measured on the ICV scale developed for rodents. **(B)** Mice (n=8) assessed in Fig. 1 were analyzed for hindlimb Grip Strength (GS) pre-dose (baseline), and 3 hours (h), 24h and 1-week (wk) post-dose. **(C)** Mice (n =8) were assessed for total activity measured by the number of beam breaks in 300 seconds in an Open Field (OF) assay at pre-dose (baseline) and 3h, 24h and 1-wk post-dose. **(D)** Hindlimb GS assessed at 3h, 24h and 1-wk post-dose in Wt mice (n =8) after increasing doses of ASO5 (30 µg, 100 µg, 300 µg, and 700 µg). **(E)** Total activity assessed by OF (n = 8) at pre-dose (baseline) and 3 hours, 24 hours and 1-week post-dose. **(F)** Representative OF traces of total activity in mice ICV injected with ASO5 in **(C)** Hindlimb GS is highly correlated between hindlimb GS ($r = -0.7611$) and OF total activity ($r = -0.8463$) with Acute Inhibition score. **(D)** Total number of Adenine (A), or Thymine (T) nucleotides in a given ASO sequence relative to the 3h Acute Inhibition score *in vivo* following 700 µg ICV injection of ASO in mice. **(E)** Number of Guanines (G) in the DNA gap vs in the 2'MOE wings nucleotides in a given ASO sequence relative to the 3h Acute Inhibition score. Stats for A performed using Two-tailed Mann Whitney test, B-E performed two-way ANOVA nonparametric Kruskal-Wallis test and Dunn's multiple comparisons post-hoc test and **(G)**: Nonparametric Spearman r correlation test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

Supplementary Figure 3



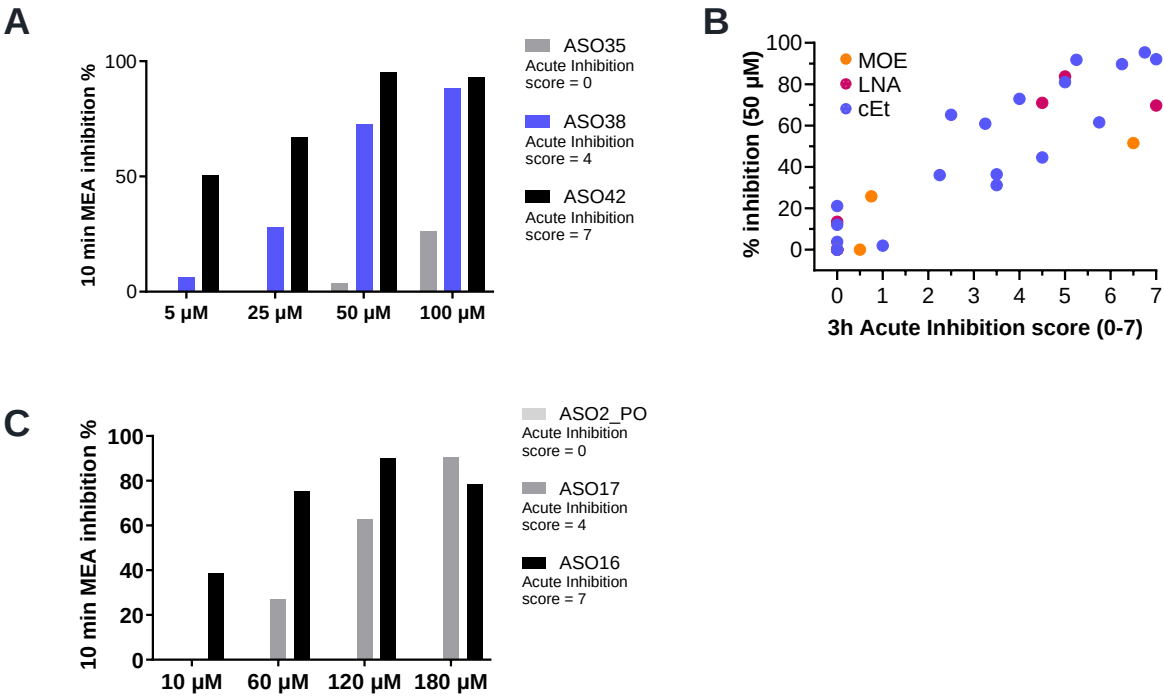
Supplementary Figure 3. Acute inhibition is sequence dependent. **(A)** Total number of Adenine (A), or Thymine (T) nucleotides in a given ASO sequence relative to the 3h Acute Inhibition score *in vivo* following 700 µg ICV injection of ASO in mice. **(B)** Number of Guanines (G) in the DNA gap vs in the 2'MOE wings nucleotides in a given ASO sequence relative to the 3h Acute Inhibition score. See Supplementary Table 9 for statistical analysis.

Supplementary Figure 4



Supplementary Figure 4. Divalent cation supplementation in ASO dosing solutions does not mitigate acute inhibition. **(A-B)** Increasing concentrations (0-100 mM) of **(A)** Mg²⁺ (blue) or **(B)** Ca²⁺ (orange) were administered in aCSF via ICV injection in mice either alone (vehicle) or with 300 µg ASO6. Acute inhibition level was assessed 3h after injection with no to minimal effect of cation supplementation on the ASO acute inhibition response (n=4 mice/group). Each data point represents individual animals' scores. *P < 0.05, **P < 0.01, ***P < 0.001, ****P < 0.0001 by the nonparametric Kruskal-Wallis test and Dunn's multiple comparisons post-hoc test. Note: x = died immediately at dosing. Multiple comparisons were done against the 0 mM cation level in each condition (+/- ASO). See Supplementary Results

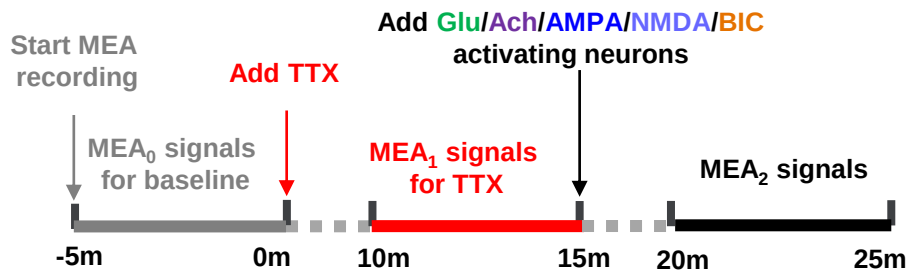
Supplementary Figure 5



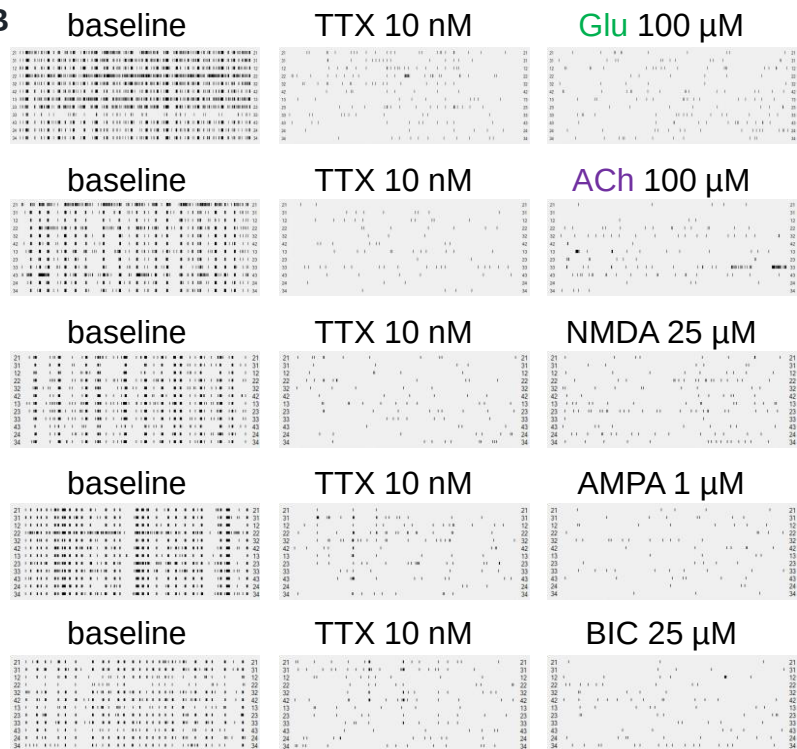
Supplementary Figure 5. *In vivo* ASO-mediated inhibition scores correlate with inhibition of neuronal firing *in vitro*. **(A)** Dose response studies with primary neurons treated with ASO35, ASO38, and ASO42 at 5 μ M, 25 μ M, 50 μ M, and 100 μ M dose use to determine the optimal ASO concentration that reflected the Acute Inhibition scores found after a 300 μ g ICV injection in mice (ASO35 = 0; ASO38 = 4; ASO42 = 7). **(B)** Percent inhibition of firing in primary neurons treated with different ASOs (LNAs = 7, cEts = 21 and MOE gapmers = 5) at a concentration of 50 μ M compared to the 3h Acute Inhibition scores ($r = 0.84$ with 95% CI = 0.69 to 0.92). **(C)** Dose response studies with primary neurons treated with ASO2_PO, ASO17, and ASO16_PO at 10 μ M, 60 μ M, 120 μ M, and 180 μ M concentration use to determine the optimal ASO concentration that reflected the Acute Inhibition scores found after a 700 μ g ICV injection in mice (ASO2_PO = 0; ASO17 = 4; ASO16 = 7).

Supplementary Figure 6

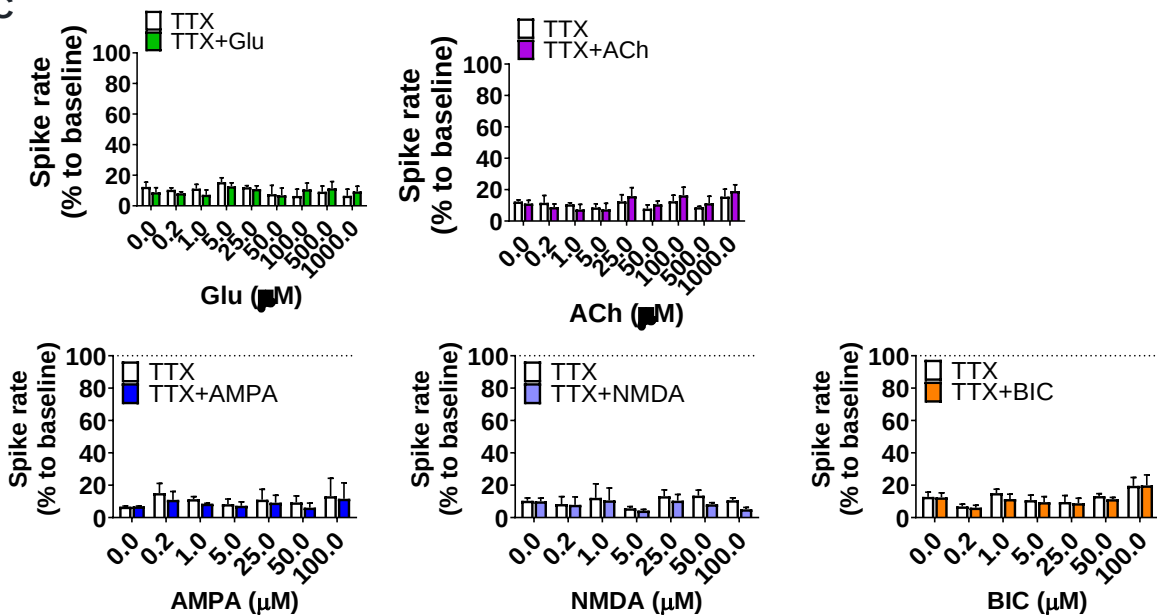
A



B



C



Supplementary Figure 6. Application of excitatory neurotransmitters cannot reverse TTX-mediated inhibition of action potential propagation. **(A)** Schematic of the experimental design with baseline recordings for 5 min, 10nM TTX addition and recording for 5 min, then exogenous neurotransmitters added in dose response with recording for an additional 5 min. **(B)** Representative raster plots of neuronal spike patterns at baseline, during treatment with TTX, and after treatment with Glu, Ach, NMDA, AMPA, or BIC. **(C)** Spike rate quantification of traces in (B) results show ~90% inhibition of neuronal firing upon treatment with TTX, similar to ASO9_PO that causes ~90% decrease in neuronal spike rates compared to baseline control. Neurotransmitter treatment with Glu, Ach, NMDA, AMPA, or BIC could not reverse TTX-mediated inhibition at any concentration. TTX = Tetrodotoxin; Glu = Glutamate; Ach = Acetylcholine; NMDA = N-methyl-D-aspartate; AMPA = α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid; Bic = Bicuculline methiodide.

Supplementary Table 1

ASOs used in *in vivo* studies

ASO #	SimpleSequence	Figure #	Supp. Fig. #
ASO1	TCTCTATTGCACATTCCAAG	1	S2A
ASO2	GTTTTCAAACACACCTTCAT	2A; 5E	S2B, S2C, S2G
ASO2_PO	GTToToToToCAAACACACCTToToCAT	5E, 6A-D	
ASO3	GTTTTCTTATTTTTAGCATT	2A; 5E	S2B, S2C, S2G
ASO3_PO	GTToToToToCTTATTTTTAGoCoATT	5E, 6A-D	
ASO4	CACTTCGAACCTCTGGCGGG	2A; 5E	S2B, S2C, S2G
ASO4_PO	CAoCoToToCGAACTCTGGGoCoGGG	5E, 6A-D	
ASO5	GCCAGGCTGGTTATGACTCA	2A, 2B, 5C,D	S2B-G
ASO5_PO	GCoCoAoGoGCTGGTTATGAoCoTCA	5C,D	
ASO6	CGAGACAGTCGCTTCCACTT	2A, 3B, 4B,4D, 5E,F, 7A-D	S2B, S2C, S2G, S4A-B
ASO6_PO (MOE)	CGoAoGoAoCAGTCGCTTCCoAoCTT	5E,F; 7A-D	
ASO6_PS_LNA	CGAGACAGTCGCTTCCACTT	5F	
ASO6_PO_LNA	CGoAoGoAoCAGTCGCTTCCoAoCTT	5F	
ASO6_PS_cEt	CGAGACAGTCGCTTCCACTT	5F	
ASO6_PO_cEt	CGoAoGoAoCAGTCGCTTCCoAoCTT	5F	
ASO7	AAGGAGGTCATCCACGAAGT	2A; 5E	S2B, S2C, S2G
ASO7_PO	AAoGoGoAoGGTCATCCACGoAoAGT	5E, 6A-D	
ASO8	TGCATGGTGTAGCCCCCTG	2A; 5E	S2B, S2C, S2G
ASO8_PO	TGoCoAoToGGTGTAGCCCCoCoCTG	5E, 6A-D	

PO = o ; orange = MOE; black = DNA; blue = cEt;
red = LNA; *all linkages are PS unless noted

Supplementary Table 2

ASOs used in *in vitro* studies

ASO #	SimpleSequence	Figure #	Supp. Fig. #
ASO2_PO	GTToToToToCAAACACACCToToCAT	8A-B; 9A-E	S5B-C
ASO3_PO	GTToToToToCTTATTTTTAGoCoATT	8A	S5B
ASO4_PO	CAoCoToToCGAACTCCTGGoCoGGG	8A	S5B
ASO5_PO	GCoCoAoGoGCTGGTTATGAoCoTCA	8A	S5B
ASO6	CGAGACAGTCGCTTCCACTT	8A	
ASO7_PO	AAoGoGoAoGGTCATCCACGoAoAGT	8A; 9A-E	S5B
ASO8_PO	TGoCoAoToGGTGTAGCCCCoCoCTG	8A-B; 9A-E	S5B
ASO9	GAoToAoToTATCCTTTGAGoCoCAC	8A	
ASO10	CCoGoToTTTCTTACC AoCoCCT	8A	
ASO11	CCoToAoTAGGACTATCCAoGoGoAA	8A	
ASO12	CAoGoAoCTGTAATCTAGGoAoCCC	8A	
ASO13	ACoGoAoCATTTCCTTGCCoToCTT	8A	
ASO14	GCoToCoATATCTAAAGACoCoGCA	8A	
ASO15	TGoAATTCCTTTACACCAoCAC	8A	
ASO16	TTToTTTGTTAATAGTTCToCTG	8A	S5C
ASO17	CGoGoToAoAACTTTATATGoGoCTC	8A	S5C
ASO18	CAoGACTGTAATCTAGGoAoCCC	8A	
ASO19	TCTCTATTGCACATTCCAAG	8A	
ASO20	GCoToToTTACTGACC AoToGCG	8A	
ASO21	CToCoToTACTCCCATCoAoCTG	8A	
ASO22	TCoToGoTCTTTGGAGCoCoCAA	8A	
ASO23	GCoToGoCGATCCCCAToToCCA	8A	
ASO24	AAoToAoCATCCATGGCoTAA	8A	
ASO25	GGoCoCoTTTGAAAGTCoCTT	8A	
ASO26	TGoToAoTTTTGGATGCoTTC	8A	
ASO27	TCoCTTTACACCAC AoCTG	8A	
ASO28	TTToToTGTTAATAGTT oCTC	8A	
ASO29	TTToToTGTTAATAGTT oCTC	8A	
ASO30	AToTCCTTTACACCAC AoCoTGG	8A	
ASO31	TTToTTTGTTAATAGTTCoToCTG	8A	
ASO32	GCoCoT TACTCTAGGAoCoCoAoAGA	8A	
ASO33	ACoCoCoTTTCCATGTGAoCoATT	8A	
ASO34	AGoCoAoATCATTGGTAGCoAoTAC	8A	

PO = o ; orange = MOE; black = DNA; blue = cEt; red = LNA; *all linkages are PS unless noted

Supplementary Table 3

ASOs used in *in vitro* studies

ASO #	SimpleSequence	Figure #	Supp. Fig. #
ASO35_LNA	ATTTCCAAATTCAC TT		S5A-B
ASO36_LNA	CTTTATTTCCAAATTCAC TT		S5B
ASO37_LNA	ATTTCCAAATTCAC TTTAC		S5B
ASO38_LNA	TAGCCCTAAAGTCC CA		S5A-B
ASO39_LNA	CATGATTGTGGGCT TA		S5B
ASO40_LNA	ACTGGTTAGCCCT AAA		S5B
ASO41_LNA	CTTTATTTCCAAATTCAC TT		S5B
ASO35_cEt	ATTTCCAAATTCAC TT		S5B
ASO36_cEt	CTTTATTTCCAAATTCAC TT		S5B
ASO37_cEt	ATTTCCAAATTCAC TTTAC		S5B
ASO38_cEt	TAGCCCTAAAGTCC CA		S5B
ASO39_cEt	CATGATTGTGGGCT TA	8D-F	S5B
ASO40_cEt	ACTGGTTAGCCCT AAA	8D,8F	S5B
ASO41_cEt	CTTTATTTCCAAATTCAC TT		S5B
ASO42	GCAAACAGGATAC AGT	8D-F	S5A-B
ASO43	CTAGCCCACCCATCAAT TTG		S5B
ASO44	GTTAGCCCTAAAGTCCC AGG		S5B
ASO45	TACATGCGTC TTT		S5B
ASO46	TTTATTTCCAAATTCAC TTT		S5B
ASO47	AAGATGAAATTTG CTC		S5B
ASO48	TACTAGCCCACCCAT C	8D,8F	S5B
ASO49	AAAGATGAAATTTGCTC TTA		S5B
ASO50	CCTTAATTTACCC CTC	8D,8F	S5B
ASO51	CTTTATTTCCAAATTCAC TT		S5B
ASO52	GA oCo TAATATGCAG ToTT		S5B
ASO53	TT oGo CCAATATCAC CoAT		S5B
ASO54	TT oGoCC AATATCAC CoAT		S5B
ASO55	CA oAoCUGA ACCACCC oGT		S5B

PO = **o** ; orange = MOE; black = DNA; blue = cEt; red = LNA; *all linkages are PS unless noted

Supplementary Table 4

Figure #	Treatment group	Number of rats	ASO (µg)	Route of delivery
1A	Saline	4	0	IT
	ASO1	4	100	IT
	ASO1	4	300	IT
	ASO1	4	700	IT
	ASO1	3	1000	IT

The heatmap shows the number of rats/group that exhibited paresis/paralysis in the designated regions over time (white = 0/4; green = 4/4). All groups were compared to a vehicle (Saline) control group.

Supplementary Table 5

Figure #	Treatment group	Number of rats	ASO (µg)	Time Point (h)	Route of delivery
1B	ASO1	3	700	0.5	IT
	ASO1	12	700	1	IT
	ASO1	18	700	3	IT
	ASO1	3	700	6	IT
	ASO1	3	700	12	IT
	ASO1	3	700	14	IT
	ASO1	3	700	16	IT
	ASO1	3	700	18	IT
	ASO1	3	700	20	IT
	ASO1	3	700	24	IT
	ASO1	3	700	420	IT
	ASO1	3	700	840	IT
1C	ASO1	7	700	3	ICV
	ASO1	7	700	24	ICV

Statistical analysis for 1C performed using Mann-Whitney two-tailed t test. The 3h vs 24h time point has a p value of 0.0006.

Supplementary Table 6

Figure #	Tx group	Number of rats	ASO (µg)	Time Point (h)	Statistical comparison	P value
1D	ICV (ASO1)	8	700	3	anterior CX vs. posterior CX	ns
					anterior CX vs. midbrain	ns
					anterior CX vs. brainstem	ns
					anterior CX vs. cervical SC	ns
					anterior CX vs. thoracic SC	ns
					anterior CX vs. lumbar SC	ns
1D	IT (ASO1)	6	700	3	anterior CX vs. posterior CX	ns
					anterior CX vs. midbrain	ns
					anterior CX vs. brainstem	ns
					anterior CX vs. cervical SC	<0.0001
					anterior CX vs. thoracic SC	<0.0001
					anterior CX vs. lumbar SC	<0.0001

Statistical analysis done using One-way ANOVA with Dunnett's multiple comparison test comparing the mean of each tissue to the anterior cortex within each group.

Supplementary Table 7

Figure #	Treatment group	Number of mice	ASO (µg)	Time Point (h)	Behavioral Test	P value
2A	PBS	16	-	3	Acute Inhibition	-
	ASO2	16	100	3	Acute Inhibition	>0.9999
	ASO3	16	100	3	Acute Inhibition	0.0487
	ASO4	16	100	3	Acute Inhibition	0.0143
	ASO5	16	100	3	Acute Inhibition	<0.0001
	ASO6	16	100	3	Acute Inhibition	<0.0001
	ASO7	16	100	3	Acute Inhibition	<0.0001
	ASO8	15	100	3	Acute Inhibition	<0.0001

Statistical analysis performed using Multiple comparisons Kruskal-Wallis test comparing ASO treated animals to the PBS control group.

Supplementary Table 8

Figure #	Treatment group	Number of mice	ASO (µg)	Time Point (h)	Behavioral Test	P value
2B	PBS	16	-	3	Acute Inhibition	-
	ASO5	16	30	3	Acute Inhibition	ns
	ASO5	16	100	3	Acute Inhibition	0.0051
	ASO5	16	300	3	Acute Inhibition	<0.0001
	ASO5	16	700	3	Acute Inhibition	<0.0001

Statistical analysis done for acute inhibition scoring using Multiple comparisons Kruskal-Wallis test comparing ASO5 to the PBS control group.

Supplementary Table 9

Figure #	Variable	ASO (µg)	Time Point (h)	By Variable	Spearman ρ
2C	G count average	700	3	Acute Inhibition	0.4795
2D	C count average	700	3	Acute Inhibition	-0.2156
S3A	A count average	700	3	Acute Inhibition	-0.1097
	T count average	700	3	Acute Inhibition	-0.0179
S3B	G count in Gap	700	3	Acute Inhibition	0.4935
	G count in Wings	700	3	Acute Inhibition	0.2052

Statistical analysis comparing the number of Guanines (Gs), Cytosines (Cs), Adenines (As), and Thymine (Ts) to the 3h Acute Inhibition score using the nonparametric Spearman r correlation test.

Supplementary Table 10

Figure #	Treatment group	Number of ASOS	ASO (µg)	Time Point (h)	Behavioral Test	P value
3A	IT vs ICV	1031	3000/700	3	Acute Inhibition	0.067

Statistical analysis comparing treatment group performed using nonparametric Spearman r correlation test comparing 3h Acute Inhibition scores for rats dosed with 3000 µg and mice dosed with 700 µg of ASO.

Supplementary Table 11

Figure #	Treatment group	Number per group	ASO (µg)	Time Point (h)	Behavioral Test	P value
3B Mouse	PBS	4	-	3	Acute Inhibition	-
	ASO6	4	30	3	Acute Inhibition	ns
	ASO6	4	100	3	Acute Inhibition	ns
	ASO6	4	300	3	Acute Inhibition	ns
	ASO6	4	700	3	Acute Inhibition	0.0016
3B Rat	PBS	4	-	3	Acute Inhibition	-
	ASO6	4	30	3	Acute Inhibition	ns
	ASO6	4	100	3	Acute Inhibition	ns
	ASO6	4	300	3	Acute Inhibition	ns
	ASO6	3	700	3	Acute Inhibition	ns
	ASO6	3	1000	3	Acute Inhibition	0.0131
3B NHP	PRE	2M/3F	-	3	Acute Inhibition	ns
	ASO6	2M/3F	30	3	Acute Inhibition	ns
	ASO6	2M/3F	100	3	Acute Inhibition	ns
	ASO6	2M/3F	300	3	Acute Inhibition	ns
	ASO6	2M/3F	700	3	Acute Inhibition	0.0043

Statistical analysis performed using Multiple comparisons Kruskal-Wallis test comparing Acute Inhibition scores for each treatment group to the PBS control group (not shown in Figure 3).

Note: PRE refers to predosing

Supplementary Table 12

Figure #	Treatment group	Number per group	ASO (μg)	Time Point (days)	Behavioral Test	P value
4B Rat	ASO6	6	100	1	3h Acute Inhibition	-
				8	3h Acute Inhibition	ns
				15	3h Acute Inhibition	ns
	ASO6	5	300	1	3h Acute Inhibition	-
				8	3h Acute Inhibition	ns
				15	3h Acute Inhibition	ns
4D NHP	ASO6	2M/3F	4000	Pre	Acute Inhibition	-
				1	Acute Inhibition	ns
				8	Acute Inhibition	ns
				15	Acute Inhibition	ns
				22	Acute Inhibition	ns
				29	Acute Inhibition	ns
				36	Acute Inhibition	ns
				43	Acute Inhibition	ns

Statistical analysis performed using Multiple comparisons Kruskal-Wallis test comparing Acute Inhibition score for each ASO dose overtime. For rats, the comparison for each dose was the 3h Acute Inhibition score on day 1. For the NHP, the comparison was compared to the pre-dose (PRE) baseline. Note: all groups were compared to a vehicle control group (data not shown).

Supplementary Table 13

Figure #	Treatment groups	Number per group (PS/PO)	ASO (μg)	Time Point (h)	Behavioral Test	P value (PS vs PO)
5C	ASO5 (PS vs PO)	4/4	10	3	Acute Inhibition	ns
		4/4	30	3	Acute Inhibition	ns
		4/4	100	3	Acute Inhibition	0.0286
		4/4	300	3	Acute Inhibition	0.0286
		4/4	700	3	Acute Inhibition	0.0286
5E	ASO2 (PS vs PO)	4/4	30	3	Acute Inhibition	ns
		4/4	100	3	Acute Inhibition	0.0286
		4/4	300	3	Acute Inhibition	0.0286
		4/4	700	3	Acute Inhibition	0.0286
	ASO3 (PS vs PO)	4/4	30	3	Acute Inhibition	ns
		4/4	100	3	Acute Inhibition	0.0286
		4/4	300	3	Acute Inhibition	0.0286
		4/4	700	3	Acute Inhibition	0.0286
	ASO4 (PS vs PO)	4/4	30	3	Acute Inhibition	ns
		4/4	100	3	Acute Inhibition	0.0286
		4/4	300	3	Acute Inhibition	0.0286
		4/4	700	3	Acute Inhibition	0.0286
	ASO6 (PS vs PO)	4/4	30	3	Acute Inhibition	ns
		4/4	100	3	Acute Inhibition	0.0286
		4/4	300	3	Acute Inhibition	0.0286
		4/4	700	3	Acute Inhibition	0.0286
	ASO7 (PS vs PO)	4/4	30	3	Acute Inhibition	ns
		4/4	100	3	Acute Inhibition	0.0286
		4/4	300	3	Acute Inhibition	0.0286
		4/4	700	3	Acute Inhibition	ns
	ASO8 (PS vs PO)	4/4	30	3	Acute Inhibition	0.0286
		4/4	100	3	Acute Inhibition	ns
		4/4	300	3	Acute Inhibition	ns
		4/4	700	3	Acute Inhibition	ns
5F-MOE	ASO6 (PS vs PO)	3/4	100	3	Acute Inhibition	0.0286
5F-LNA	ASO6 (PS vs PO)	4/4	100	3	Acute Inhibition	0.0286
5F-cEt	ASO6 (PS vs PO)	4/4	100	3	Acute Inhibition	0.0286

Statistical analysis comparing Acute Inhibition scores for ASOs with the same sequence but different chemical modifications at escalating doses perform with the nonparametric Mann-Whitney test.

Note: ASO# = full PS modified ASOs and ASO#_PO = mixed PS and PO modified ASOs (see Supplementary Table 1 for sequence info)

Supplementary Table 14

Figure #	Treatment group	Number per group	ASO (μ g)	Time Point (h)	P value (Pre vs 2h)
6A	ASO2_PO	3	700	2	ns
	ASO3_PO	3	700	2	ns
	ASO4_PO	2	700	2	0.0068
	ASO7_PO	2	700	2	0.0021
	ASO8_PO	3	700	2	<0.0001
6B	ASO2_PO	3	700	2	0.0021
	ASO3_PO	3	700	2	ns
	ASO4_PO	2	700	2	0.0011
	ASO7_PO	2	700	2	<0.0001
	ASO8_PO	3	700	2	<0.0001
6C	ASO2_PO	3	700	2	ns
	ASO3_PO	3	700	2	ns
	ASO4_PO	2	700	2	ns
	ASO7_PO	2	700	2	ns
	ASO8_PO	3	700	2	0.0246
6D	ASO2_PO	3	700	2	ns
	ASO3_PO	3	700	2	ns
	ASO4_PO	2	700	2	ns
	ASO7_PO	2	700	2	ns
	ASO8_PO	3	700	2	ns

Statistical analysis performed using Two-way ANVOA comparing baseline (Pre) recordings to the 2 hour (h) time point within each group. Note: Baseline (Pre) recordings compared to 24h time point are all nonsignificant (ns).

Supplementary Table 15

Figure #	Treatment group	Number of rats	ASO (μg)	Time Point (h)	P value
7A	ASO6 (PS vs PO)	4/4	700	1	<0.0001
		4/4	700	3	<0.0001
		4/4	700	6	<0.0001
		4/4	700	24	ns
7B	ASO6 (PS vs PO)	4/3	700	1	ns
		3/3	700	3	ns
		4/4	700	6	ns
		3/3	700	24	0.0487

Figure 7A the statistical analysis for acute inhibition scoring performed using the nonparametric Mann-Whitney test comparing PS vs PO ASO at each time point.

Figure 7B, the statistical analysis for ASO concentration was Two-way ANOVA Mixed-effects analysis using Two-way ANOVA Šídák's post hoc

Supplementary Table 16

Figure #	Study	Number of ASOS	ASO (μM)	r	95% CI
S5B	correlation	33	50	0.84	0.79 to 0.95
8A	correlation	34	120	0.9	0.83 to 0.98

Statistical analysis comparing in vivo acute inhibition to in vitro % inhibition of neuronal firing performed using nonparametric Spearman r correlation test

Supplementary Table 17

Figure #	Treatment group	# of wells	ASO (μM)	Washout Time Point (m/h)	Change in Spike rate +/- Tx (P value)
8F	ASO50	24	120	10 m	ns
	ASO48	36	120	10 m	0.0001
	ASO40	36	120	10 m	<0.0001
	ASO42	36	120	10 m	<0.0001
	ASO50	24	120	1 h	ns
	ASO48	36	120	1 h	0.0039
	ASO40	36	120	1 h	<0.0001
	ASO42	36	120	1 h	<0.0001
	ASO50	24	120	2 h	ns
	ASO48	36	120	2 h	0.0014
	ASO40	36	120	2 h	<0.0001
	ASO42	36	120	2 h	<0.0001

Statistical analysis performed was Two-way ANOVA Šídák's post hoc test comparing spike rate with ASO Tx to spike rate following washout at either 10min, 1h, or 2h.

Tx = treatment

Supplementary Table 18

Figure #	ASO	# of wells (+/- Tx)	Tx group	Tx concentration (μ M)	Change in Spike rate +/- Tx (P value)
9A	ASO8_PO	3/3	Glutamate	0	ns
		3/3	Glutamate	0.2	ns
		3/3	Glutamate	1	ns
		3/3	Glutamate	5	0.0022
		3/3	Glutamate	25	<0.0001
		3/3	Glutamate	50	<0.0001
		3/3	Glutamate	100	<0.0001
		3/3	Glutamate	500	ns
		3/3	Glutamate	1000	ns
	ASO7_PO	3/3	Glutamate	0	ns
		3/3	Glutamate	0.2	ns
		3/3	Glutamate	1	ns
		3/3	Glutamate	5	ns
		3/3	Glutamate	25	0.0468
		3/3	Glutamate	50	<0.0001
		3/3	Glutamate	100	<0.0001
		3/3	Glutamate	500	ns
		3/3	Glutamate	1000	ns
	ASO2_PO	3/3	Glutamate	0	ns
		3/3	Glutamate	0.2	ns
		3/3	Glutamate	1	ns
		3/3	Glutamate	5	ns
		3/3	Glutamate	25	ns
		3/3	Glutamate	50	0.0012
		3/3	Glutamate	100	0.0247
		3/3	Glutamate	500	ns
		3/3	Glutamate	1000	ns

Statistical analysis performed was Two-way ANOVA Šídák's post hoc test comparing +/- Tx (increasing concentrations of Glutamate) to ASO (120 μ M).

Tx = treatment

Supplementary Table 19

Figure #	ASO	# of wells (+/- Tx)	Tx group	Tx concentration (μM)	Change in Spike rate +/- Tx (P value)
9B	ASO8_PO	3/3	Acetylcholine	0	ns
		3/3	Acetylcholine	0.2	ns
		3/3	Acetylcholine	1	ns
		3/3	Acetylcholine	5	ns
		3/3	Acetylcholine	25	0.0032
		3/3	Acetylcholine	50	<0.0001
		3/3	Acetylcholine	100	<0.0001
		3/3	Acetylcholine	500	0.0166
		3/3	Acetylcholine	1000	ns
	ASO7_PO	3/3	Acetylcholine	0	ns
		3/3	Acetylcholine	0.2	ns
		3/3	Acetylcholine	1	ns
		3/3	Acetylcholine	5	ns
		3/3	Acetylcholine	25	0.0276
		3/3	Acetylcholine	50	<0.0001
		3/3	Acetylcholine	100	0.0022
		3/3	Acetylcholine	500	ns
		3/3	Acetylcholine	1000	ns
	ASO2_PO	3/3	Acetylcholine	0	ns
		3/3	Acetylcholine	0.2	ns
		3/3	Acetylcholine	1	ns
		3/3	Acetylcholine	5	ns
		3/3	Acetylcholine	25	ns
		3/3	Acetylcholine	50	0.0036
		3/3	Acetylcholine	100	0.0002
		3/3	Acetylcholine	500	ns
		3/3	Acetylcholine	1000	ns

Statistical analysis performed was Two-way ANOVA Šídák's post hoc test comparing +/- Tx (increasing concentrations of Acetylcholine) to ASO (120 μM).

Tx = treatment

Supplementary Table 20

Figure #	ASO	# of wells (+/- Tx)	Tx group	Tx concentration (μM)	Change in Spike rate +/- Tx (P value)
9C	ASO8_PO	3/3	NMDA	0	ns
		3/3	NMDA	0.2	ns
		3/3	NMDA	1	ns
		3/3	NMDA	5	0.0406
		3/3	NMDA	25	ns
		3/3	NMDA	50	ns
		3/3	NMDA	100	ns
	ASO7_PO	3/3	NMDA	0	ns
		3/3	NMDA	0.2	ns
		3/3	NMDA	1	ns
		3/3	NMDA	5	0.0016
		3/3	NMDA	25	<0.0001
		3/3	NMDA	50	ns
		3/3	NMDA	100	0.0082
	ASO2_PO	3/3	NMDA	0	ns
		3/3	NMDA	0.2	ns
		3/3	NMDA	1	ns
		3/3	NMDA	5	0.0290
		3/3	NMDA	25	0.0030
		3/3	NMDA	50	ns
		3/3	NMDA	100	0.0001
Statistical analysis performed was Two-way ANOVA Šídák's post hoc test comparing +/- Tx (increasing concentrations of NMDA) to ASO (120 μM).					
Tx = treatment					

Supplementary Table 21

Figure #	ASO	# of wells (+/- Tx)	Tx group	Tx concentration (μ M)	Change in Spike rate +/- Tx (P value)
9D	ASO8_PO	3/3	AMPA	0	ns
		3/3	AMPA	0.2	ns
		3/3	AMPA	1	0.0059
		3/3	AMPA	5	ns
		3/3	AMPA	25	ns
		2/2	AMPA	50	ns
		3/3	AMPA	100	ns
	ASO7_PO	3/3	AMPA	0	ns
		3/3	AMPA	0.2	ns
		3/3	AMPA	1	0.0010
		3/3	AMPA	5	ns
		3/3	AMPA	25	ns
		3/3	AMPA	50	ns
		3/3	AMPA	100	0.0403
	ASO2_PO	3/3	AMPA	0	ns
		3/3	AMPA	0.2	ns
		3/3	AMPA	1	<0.0001
		3/3	AMPA	5	0.0003
		3/3	AMPA	25	<0.0001
		3/3	AMPA	50	<0.0001
		3/3	AMPA	100	<0.0001

Statistical analysis performed was Two-way ANOVA Šídák's post hoc test comparing +/- Tx (increasing concentrations of AMPA) to ASO (120 μ M).

Tx = treatment

Supplementary Table 22

Figure #	ASO	# of wells (+/- Tx)	Tx group	Tx concentration (μ M)	Change in Spike rate +/- Tx (P value)
9E	ASO8_PO	3/3	Bicuculline	0	ns
		3/3	Bicuculline	0.2	ns
		3/3	Bicuculline	1	ns
		3/3	Bicuculline	5	0.0318
		3/3	Bicuculline	25	ns
		3/3	Bicuculline	50	ns
		3/3	Bicuculline	100	ns
	ASO7_PO	3/3	Bicuculline	0	ns
		3/3	Bicuculline	0.2	ns
		3/3	Bicuculline	1	ns
		3/3	Bicuculline	5	ns
		3/3	Bicuculline	25	0.0021
		3/3	Bicuculline	50	0.0186
		3/3	Bicuculline	100	0.0216
	ASO2_PO	3/3	Bicuculline	0	ns
		3/3	Bicuculline	0.2	ns
		3/3	Bicuculline	1	ns
		3/3	Bicuculline	5	ns
		3/3	Bicuculline	25	0.0211
		3/3	Bicuculline	50	<0.0001
		3/3	Bicuculline	100	0.0001

Statistical analysis performed was Two-way ANOVA Šídák's post hoc test comparing +/- Tx (increasing concentrations of Bicuculline) to ASO (120 μ M).
Tx = treatment